Session 7 Mini Challenge: Unfreeze More and Compare

Computer Vision and Speech Recognition — Slide 29

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Repository: https://github.com/KacperDudzik/CV_session

Code: session7/session7.py (run with `python session7.py mini-challenge`)

Raw results: _output/mini_challenge_results.json

1. Task (slide 29)

Start from a fresh pretrained ResNet-18. Train with **layer3 + layer4 + fc** unfrozen at lr = 0.0001, same number of epochs as the **layer4 + fc** baseline (6 epochs, seed 0). Measure trainable parameters, seconds per epoch, and best **validation** macro-F1. Decide whether unfreezing layer3 is worth it. The test set stays closed.

2. Method

Both runs use the same lesion-grouped dermoscopic split, ImageNet normalization, random flips, class-weighted cross-entropy, Adam, and early selection by best validation macro-F1. Freezing follows the slide hint: `requires_grad` only for names starting with the chosen blocks; the optimizer is built on that trainable list only.

3. Results

Setting	Trainable params	s / epoch	Best epoch	Best val macro-F1
layer4 + fc	8,395,267	98	3	0.645
layer3 + layer4 + fc	10,494,979	123	3	0.660

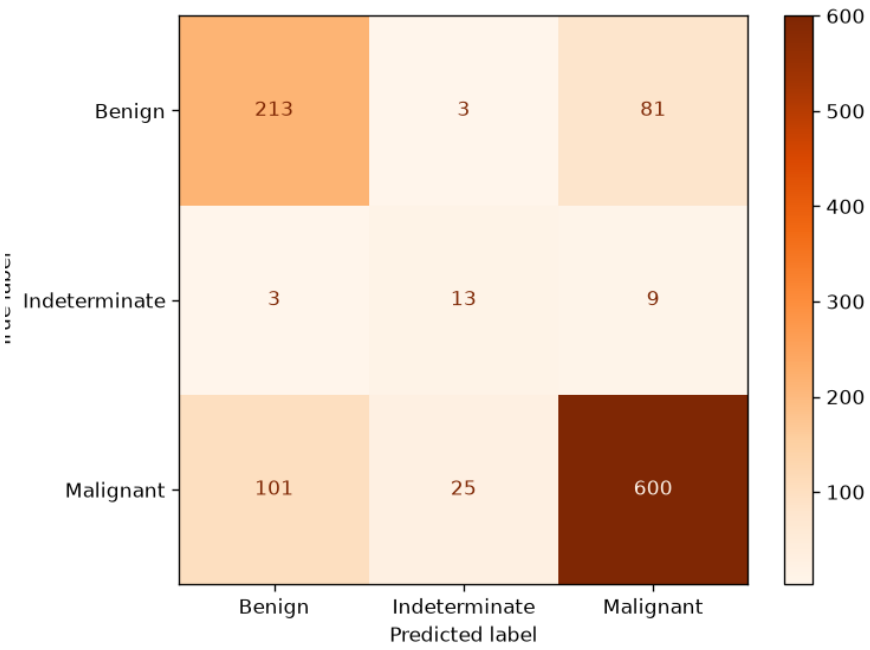
Delta: F1 **+0.015**, time **+25 s/epoch** (about 25% slower).

4. Decision

Worth it: layer3+layer4 improves validation macro-F1 enough to justify the cost.

Unfreezing layer3 trains about 2.1M more weights (10.49M vs 8.40M) and costs ~25 s more per epoch on CPU. Best validation macro-F1 rises from 0.645 to 0.660. On this run the gain is modest but positive and reappears at the same best epoch (3), so for this dataset size it is a reasonable trade-off if compute budget allows. The test set was not used.

Confusion: layer4 + fc (best epoch)



Confusion: layer3 + layer4 + fc (best epoch)

